LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	18
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Pneumatic system basic design requirements, brief description, advantages and disadvantages of a typical pneumatic system.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Pneumatic system basic design requirements, brief description, advantages and disadvantages of a typical pneumatic system.
5. Specific Objectives	23.1.4.1 Trainee will be able to describe the construction and functioning of a typical pneumatic system and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of system diagrams; Safety: No live pressure; Instructor supervision: Monitor diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary working fluid in a pneumatic system?
(b) Name one major advantage of pneumatic systems over hydraulic systems.
(c) What is the basic design requirement regarding air cleanliness in a pneumatic system?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic design requirements for pneumatic systems • Brief description of typical pneumatic system layout • Functioning of air compressor and storage components • Role of control valves and actuators in system • Standard safety features in pneumatic design	L & Ds	PPT, IB	15 min
(b)	• Advantages of pneumatic systems in aircraft • Disadvantages and limitations of pneumatic systems • Comparison with hydraulic and electrical systems • Impact of air compressibility on system response • Maintenance considerations for pneumatic components	L & Ds	PPT, PDF	20 min
(c)	• Summary of typical pneumatic system construction • Key functional elements and their interactions • Review of design requirements and constraints • Identification of common system failure points • Preparation for next lesson on components	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the compressibility of air affects the response time of a pneumatic actuator.
(b) List two specific advantages and two specific disadvantages of using a pneumatic system in an aircraft environment.

12. Home Assignment and Follow Up	Prepare a brief note on the specific components to be covered in LP 19, focusing on their individual purposes within the typical pneumatic system.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Description purpose typical pneumatic system components	Refer to the master syllabus reference list	LP 19

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min